

Volume Volatility Backtest

2,367 tickers · daily bars 2005–2026 · signals scored 2009-01 to 2026-10 · forward 7 / 14 / 21 / 42 sessions

Bottom line

- **High volume volatility predicts negative forward risk-adjusted returns.** 20-session window, ranked against each ticker's own past year. Down days: Q5–Q1 -0.045 to -0.025 (t -4.7 to -3.7). Up days: -0.037 to -0.011 (t -3.9 to -1.8).
- **Best bucket: down day + calmest volume (Q1),** $+0.017$ to $+0.024$ (t $+3.4$ to $+4.1$). Raw 21d forward return 1.78% for Q1 vs 1.55% for Q5 on down days.
- **Worst bucket: up day + most erratic volume (Q5),** -0.024 to -0.013 (t -3.7 to -3.2).
- **Every year.** The 20-session down-day spread is negative in 17 of 17 full years (2009–2025); up days 16 of 17.
- **Window: 20 sessions is the best setting over 2009–26** (avg t -3.4 ; strongest of the 15 tested, 5 to 252). Windows from 10 to 50 all work. The 60–252 windows only worked in 2022–26. 252 is the weakest (avg t -1.0).
- **Strongest combination: 20-session level + fast ROC 5/5.** Long calm volume (Q1) with a fresh short-term jump (ROC Q5); short erratic volume (Q5) with short-term volume volatility falling (ROC Q1). Average long-short $+0.038$ vs $+0.027$ for the level alone ($+38\%$), t $+3.7$. Bigger spread than the level alone in 3 of 3 periods; higher t in 2 of 3.
- **Rate of change, slow version: nothing durable.** With the volume-volatility level held fixed, 0 of 180 window $\times$ lag settings stay negative ($t \leq -1$) in all three periods. The best ones worked in 2009–2022 and went flat in 2022–26.
- **Rate of change, fast version: a jump predicts outperformance.** At a 5–15 session window and 3–5 session lag, a short-term rise in volume volatility at a given level is positive in all three periods. Full-sample t $+2.1$ to $+2.6$. That runs opposite to the level effect.

How to read the numbers

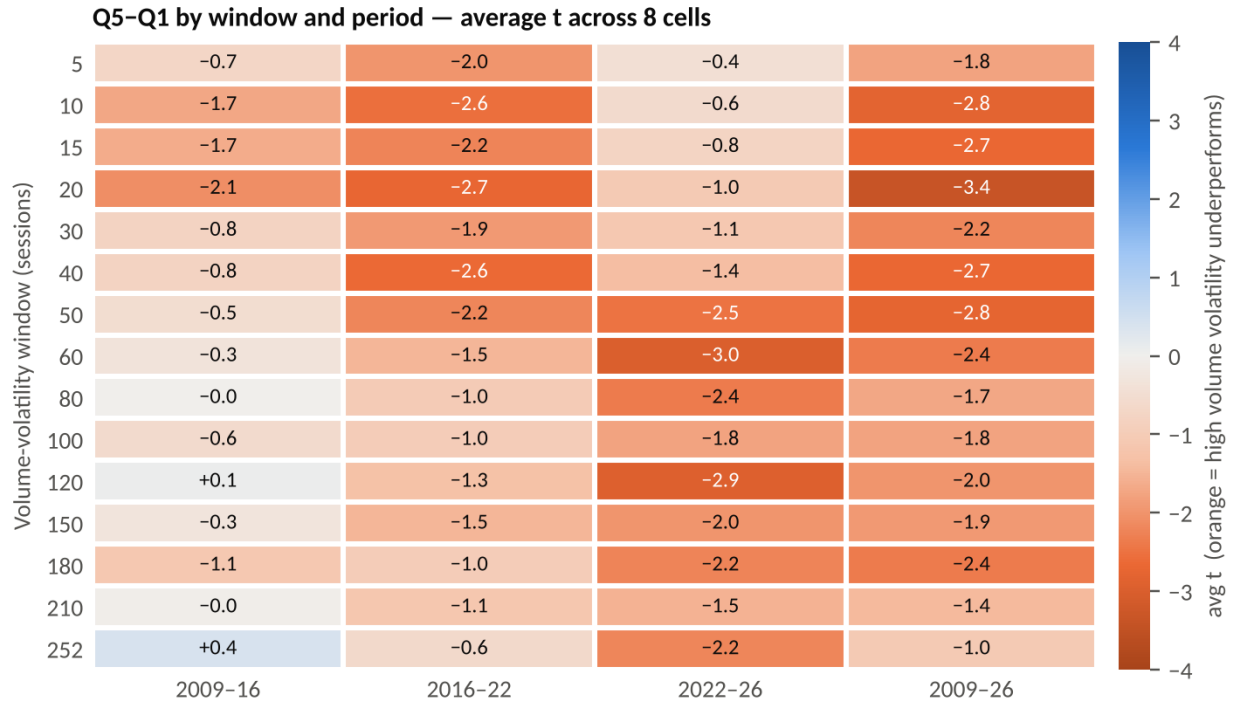
Each figure is the forward return divided by the stock's trailing 20-day volatility scaled to the horizon ($\sigma \cdot \sqrt{h}$), minus that day's universe average. 0 = in line with the universe; $-0.030 = 3\%$ of a horizon- σ worse. Q1–Q5 are fifths of each ticker's own trailing-252-session distribution of the factor. Brackets hold Newey-West t -stats (lag = horizon). Each bucket is averaged across names per date, then across dates.

Setup

Universe	2,367 tickers from the finviz export (today's list); a name is scored once it has 756 sessions of history, so every setting is tested on the same names
Data	Yahoo daily OHLCV via yfinance, split/dividend adjusted, 2005-01 to 2026-10. Volume confirmed split-adjusted
Volume volatility	Stdev of daily $\ln(V_t/V_{t-1})$ over N sessions. Bad prints ($<2\%$ of 20d median) dropped; daily change capped at $50\times$
Rate of change	$VV_t / VV_{t-L} - 1$
Day type	Up: close $>$ prior close. Down: close $<$ prior close
Entry / filter	Close of signal day. Price $\geq \$1$, 20d median dollar volume $\geq \$500k$
Periods	2009-01 to 2016-09 (never used in earlier rounds) · 2016-10 to 2022-06 · 2022-07 to 2026-10

1. Window length: 5 to 252 sessions

Fifteen windows, each scored over three periods on the same dates and names. The rate of change in Section 3 uses the same grid.

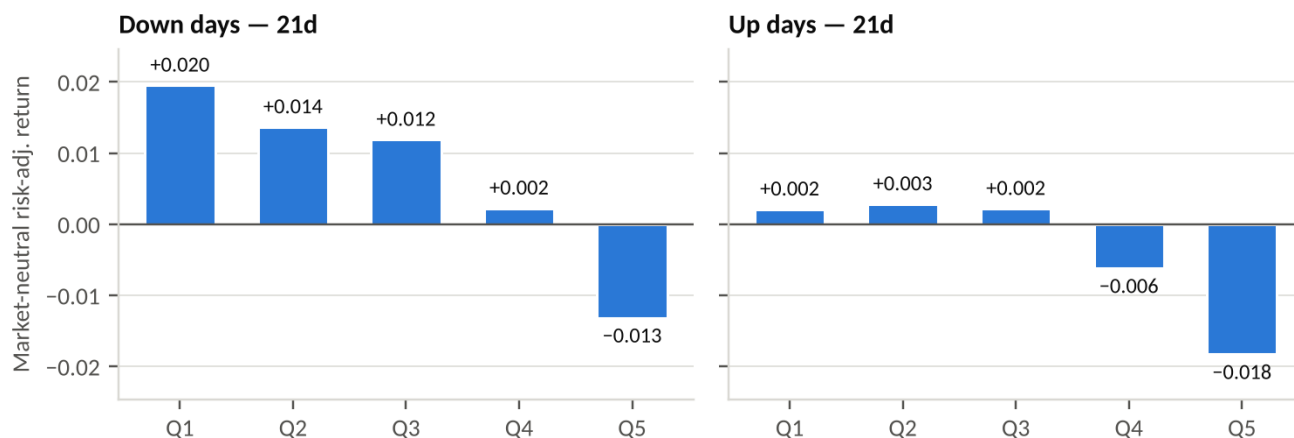


- 2009-16, never examined before this round: 10-20 sessions work (20: avg t -2.1). 40 sessions and longer show nothing.
- 2016-22: 10-50 work. 2022-26: 50-180 work best, short windows fade.
- The ranking of windows flips between periods: rank correlation of window avg t, 2009-16 vs 2022-26 = -0.66, 2016-22 vs 2022-26 = -0.54. The 40-60 call from the 2018-26 round was a 2022-26 effect.
- Over the full 2009-26 sample the strongest window is 20 (avg t -3.4).

Full-sample Q5-Q1 by cell, selected windows (2009-26)

Window — Down days	7d	14d	21d	42d
5 sessions	-0.011 (-2.7)	-0.012 (-2.8)	-0.011 (-2.4)	-0.012 (-2.4)
10 sessions	-0.015 (-2.9)	-0.020 (-3.3)	-0.021 (-3.1)	-0.027 (-3.7)
20 sessions	-0.025 (-4.2)	-0.030 (-3.9)	-0.032 (-3.7)	-0.045 (-4.7)
40 sessions	-0.012 (-2.5)	-0.020 (-3.0)	-0.024 (-3.2)	-0.033 (-3.8)
60 sessions	-0.010 (-2.2)	-0.017 (-3.1)	-0.017 (-2.7)	-0.020 (-2.4)
120 sessions	-0.008 (-1.9)	-0.012 (-2.3)	-0.013 (-2.1)	-0.016 (-1.9)
252 sessions	-0.002 (-0.4)	-0.005 (-0.9)	-0.004 (-0.7)	-0.010 (-1.3)
Window — Up days	7d	14d	21d	42d
5 sessions	+0.001 (+0.3)	-0.003 (-0.7)	-0.005 (-1.1)	-0.011 (-2.3)
10 sessions	-0.004 (-0.9)	-0.014 (-2.3)	-0.017 (-2.6)	-0.024 (-3.7)
20 sessions	-0.011 (-1.8)	-0.019 (-2.4)	-0.021 (-2.4)	-0.037 (-3.9)
40 sessions	-0.007 (-1.3)	-0.015 (-2.1)	-0.018 (-2.3)	-0.028 (-3.4)
60 sessions	-0.009 (-2.1)	-0.014 (-2.4)	-0.015 (-2.2)	-0.017 (-1.9)
120 sessions	-0.006 (-1.5)	-0.011 (-2.0)	-0.013 (-2.1)	-0.017 (-1.9)
252 sessions	-0.003 (-0.9)	-0.005 (-0.9)	-0.009 (-1.4)	-0.015 (-1.8)

2. The 20-session window in detail (2009–26)

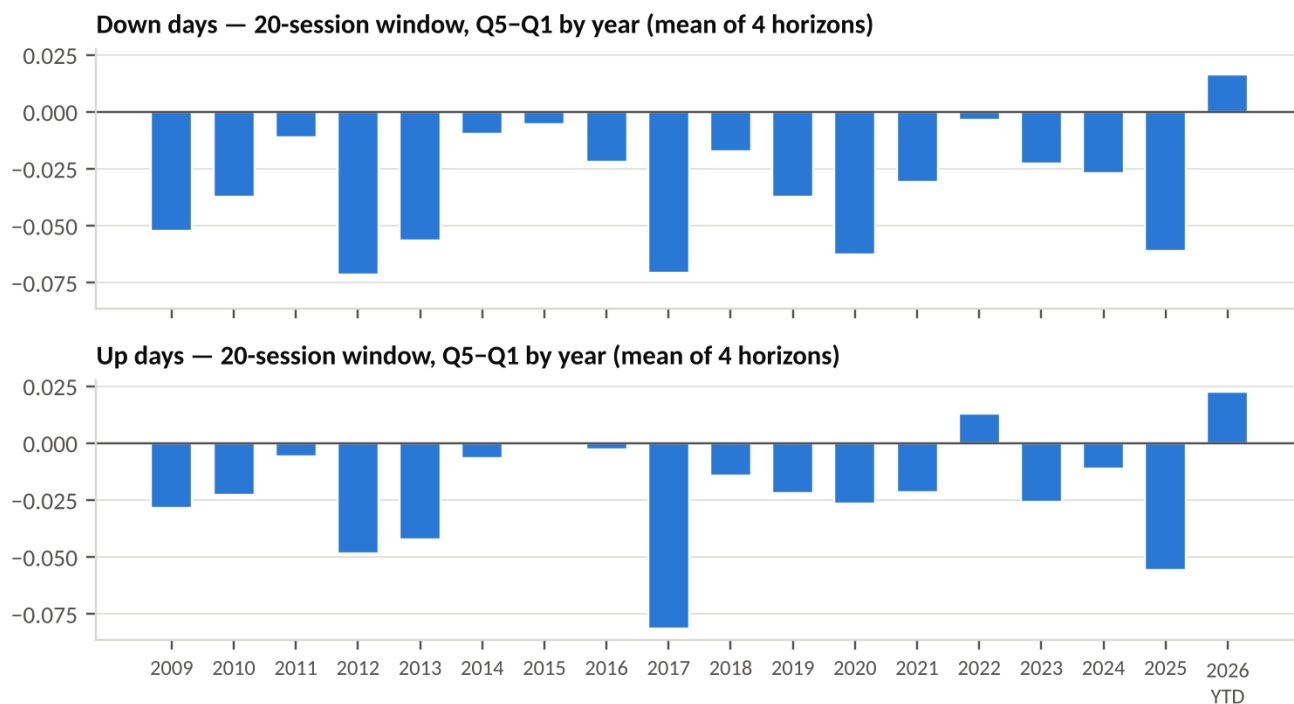


Q1 = calmest volume vs the ticker's own past year · Q5 = most erratic

Down days	7d	14d	21d	42d
Q1	+0.017 (+4.0)	+0.018 (+3.5)	+0.020 (+3.4)	+0.024 (+4.1)
Q2	+0.012 (+3.9)	+0.013 (+3.8)	+0.014 (+3.5)	+0.013 (+2.6)
Q3	+0.011 (+4.0)	+0.012 (+3.7)	+0.012 (+3.5)	+0.007 (+2.0)
Q4	+0.003 (+1.2)	+0.001 (+0.4)	+0.002 (+0.6)	+0.000 (+0.1)
Q5	-0.008 (-2.2)	-0.011 (-2.6)	-0.013 (-2.6)	-0.021 (-3.4)
All	+0.007 (+3.9)	+0.006 (+3.2)	+0.005 (+2.7)	+0.005 (+2.1)
Q5-Q1	-0.025 (-4.2)	-0.030 (-3.9)	-0.032 (-3.7)	-0.045 (-4.7)
Up days	7d	14d	21d	42d
Q1	-0.002 (-0.6)	+0.003 (+0.5)	+0.002 (+0.4)	+0.013 (+2.2)
Q2	+0.001 (+0.2)	+0.002 (+0.7)	+0.003 (+0.8)	+0.002 (+0.5)
Q3	+0.001 (+0.4)	+0.002 (+0.6)	+0.002 (+0.6)	+0.001 (+0.2)
Q4	-0.007 (-2.2)	-0.007 (-2.0)	-0.006 (-1.6)	-0.006 (-1.4)
Q5	-0.013 (-3.3)	-0.016 (-3.2)	-0.018 (-3.2)	-0.024 (-3.7)
All	-0.004 (-2.1)	-0.004 (-1.6)	-0.004 (-1.6)	-0.003 (-1.0)
Q5-Q1	-0.011 (-1.8)	-0.019 (-2.4)	-0.021 (-2.4)	-0.037 (-3.9)

- Down days: Q1 through Q5 step down in order at every horizon. The calmest-volume Q1 bucket is positive with t +3.4 to +4.1.
- Up days: the spread builds with horizon and is strongest at 42d.

2b. Year by year



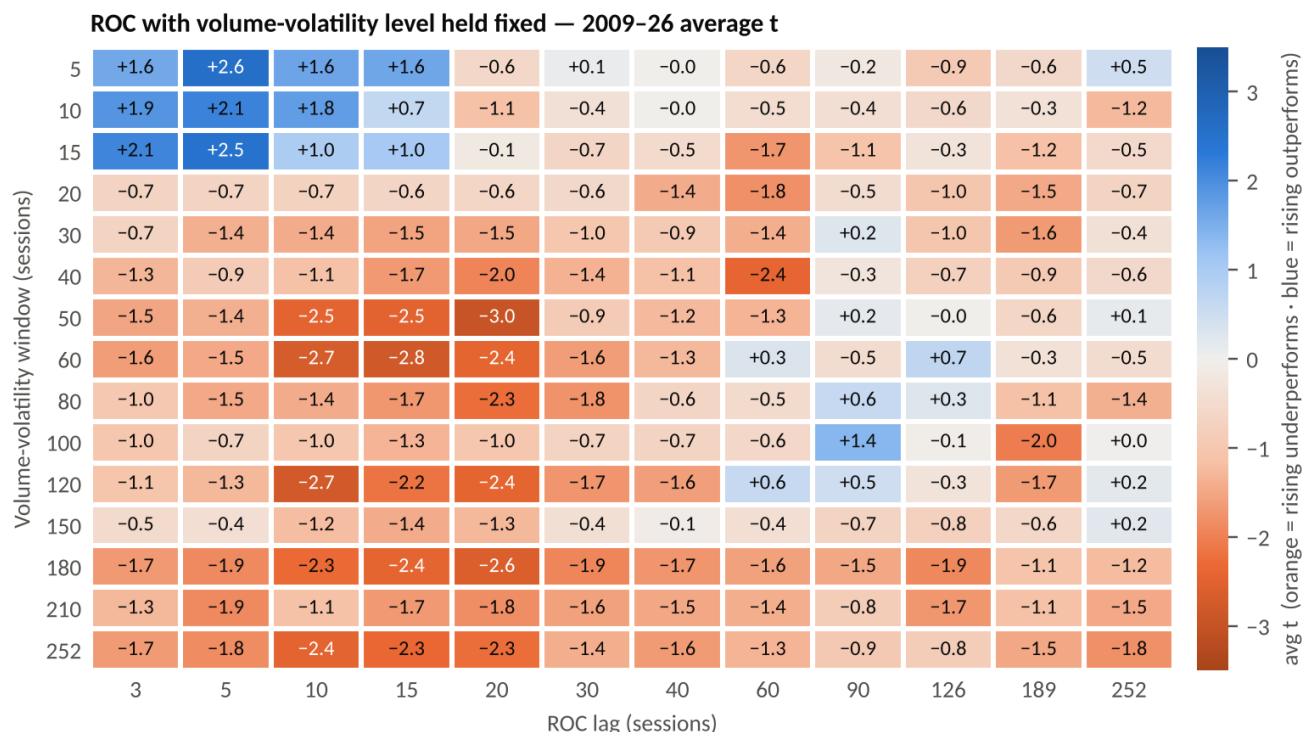
- 20-session window: down-day spread negative in 17/17 full years (2009–2025), up-day 16/17. 2026 YTD is positive.

Q5–Q1 by year and window (mean of 4 horizons)

Year	20 • Down	20 • Up	60 • Down	60 • Up	252 • Down	252 • Up
2009	-0.052	-0.029	-0.004	+0.017	+0.008	+0.019
2010	-0.037	-0.023	+0.005	+0.031	+0.057	+0.044
2011	-0.011	-0.006	+0.006	-0.019	+0.009	+0.009
2012	-0.072	-0.048	-0.055	-0.040	-0.018	-0.009
2013	-0.057	-0.042	-0.004	-0.010	+0.003	-0.010
2014	-0.010	-0.006	+0.003	-0.017	-0.018	-0.035
2015	-0.005	-0.000	-0.027	-0.016	-0.019	-0.021
2016	-0.022	-0.003	+0.038	+0.034	-0.001	-0.009
2017	-0.070	-0.081	-0.020	-0.039	-0.016	-0.012
2018	-0.017	-0.014	-0.008	-0.028	-0.027	-0.033
2019	-0.037	-0.022	-0.007	+0.002	+0.015	+0.024
2020	-0.062	-0.027	-0.010	-0.005	+0.016	+0.019
2021	-0.031	-0.022	-0.020	-0.019	+0.002	-0.010
2022	-0.003	+0.013	-0.020	-0.003	-0.008	-0.021
2023	-0.023	-0.026	-0.042	-0.028	-0.002	-0.003
2024	-0.027	-0.011	-0.047	-0.047	-0.036	-0.025
2025	-0.061	-0.056	-0.048	-0.034	-0.026	-0.035
2026 YTD	+0.017	+0.022	-0.030	-0.033	-0.040	-0.043
Negative (of 17)	17	16	13	13	10	12

3. Rate of change in volume volatility

Without holding the level fixed, ROC mostly re-measures the level: when volume volatility is rising, it is usually already high. The test that matters is ROC's Q5-Q1 inside each third of volume-volatility level, averaged. That is ROC's independent signal.



Strongest settings, level held fixed (avg t by period)

Window / lag	2009-16	2016-22	2022-26	2009-26
50 / 20	-2.4	-2.4	+0.0	-3.0
60 / 15	-2.2	-2.0	-0.4	-2.8
120 / 10	-2.3	-2.6	+0.4	-2.7
60 / 10	-2.0	-1.9	-0.5	-2.7
180 / 20	-2.2	-1.5	-0.7	-2.6
50 / 15	-2.1	-2.2	+0.3	-2.5

- Slow ROC (orange block, window 50-120, lag 10-20): worked in 2009-16 and 2016-22 at $t \approx -2$, then went flat or positive in 2022-26. Settings with $t \leq -1$ in all three periods: 0 of 180. The 60/10 setting from the last round follows that pattern.

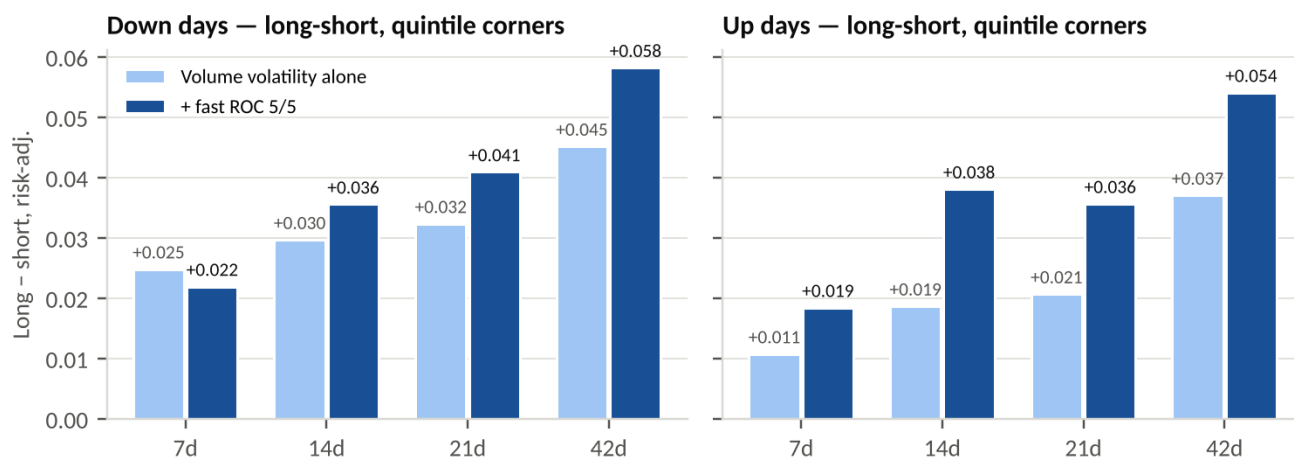
Fast ROC, level held fixed (avg t by period)

Window / lag	2009-16	2016-22	2022-26	2009-26
5 / 5	+1.3	+2.3	+0.9	+2.6
10 / 5	+1.2	+0.7	+1.9	+2.1
15 / 3	+1.4	+0.6	+1.8	+2.1
15 / 5	+0.9	+1.5	+2.2	+2.5

- Fast ROC (blue block, window 5-15, lag 3-15): positive spread in all three periods for 10 of 12 settings. Positive Q5-Q1 means a recent jump in volume volatility outperforms a recent drop at the same level.
- Weaker than the level effect: per-period t mostly +0.6 to +2.3, full-sample +2.1 to +2.6 on the best settings. It points the other way from the level, so the two are separate signals.

4. Combinations — 20-session level × ROC

Each ROC setting is crossed with the 20-session level, by day type. The long-short takes the quintile corners, with each side set by the factor's sign. Level: low is long. Fast ROC (window ≤ 15): high is long. Slower ROC: low is long. Positive = the combination works. All figures 2009–26, same universe as the sweep.



Ranking, average of 8 day × horizon cells

Signal	Avg long-short	vs level alone	Avg t	2009–16 t	2016–22 t	2022–26 t
Level alone	+0.027	+0%	+3.4	+2.1	+2.7	+1.0
Level + ROC 20/10	+0.035	+27%	+3.7	+2.9	+2.8	+0.7
Level + ROC 5/5 (fast)	+0.038	+38%	+3.7	+2.4	+2.5	+1.5
Level + ROC 10/5 (fast)	+0.038	+37%	+3.4	+2.4	+1.3	+2.4
Level + ROC 120/20	+0.031	+13%	+3.4	+2.0	+2.4	+1.4
Level + ROC 50/20	+0.032	+18%	+3.3	+2.0	+2.7	+1.1
Level + ROC 15/5 (fast)	+0.035	+28%	+3.3	+1.6	+2.5	+1.7
Level + ROC 15/3 (fast)	+0.032	+18%	+3.2	+1.6	+2.1	+1.8

Long-short by horizon, quintile corners

Signal — Down days	7d	14d	21d	42d
Level alone	+0.025 (+4.2)	+0.030 (+3.9)	+0.032 (+3.7)	+0.045 (+4.7)
Level + ROC 5/5	+0.022 (+2.6)	+0.036 (+3.5)	+0.041 (+3.7)	+0.058 (+5.2)
Level + ROC 10/5	+0.033 (+3.8)	+0.039 (+3.7)	+0.041 (+3.5)	+0.063 (+5.1)
Level + ROC 15/5	+0.020 (+2.3)	+0.031 (+3.0)	+0.034 (+3.0)	+0.055 (+5.0)
Level + ROC 15/3	+0.024 (+2.9)	+0.024 (+2.4)	+0.028 (+2.5)	+0.053 (+4.9)
Signal — Up days	7d	14d	21d	42d
Level alone	+0.011 (+1.8)	+0.019 (+2.4)	+0.021 (+2.4)	+0.037 (+3.9)
Level + ROC 5/5	+0.019 (+2.2)	+0.038 (+3.8)	+0.036 (+3.5)	+0.054 (+4.7)
Level + ROC 10/5	+0.017 (+1.9)	+0.027 (+2.5)	+0.028 (+2.4)	+0.053 (+4.5)
Level + ROC 15/5	+0.021 (+2.4)	+0.031 (+3.0)	+0.033 (+3.0)	+0.057 (+4.7)
Level + ROC 15/3	+0.020 (+2.5)	+0.027 (+2.8)	+0.032 (+3.0)	+0.050 (+4.4)

Strongest single buckets (vs the universe)

Day	Horizon	Level	ROC	vs universe	Names/day
Up	42d	Q5	20/10 Q5	-0.034 (-4.8)	62
Down	42d	Q5	10/5 Q1	-0.038 (-4.7)	36
Down	42d	Q1	15/5 Q5	+0.031 (+4.7)	29
Up	42d	Q5	10/5 Q5	-0.031 (-4.5)	39
Down	42d	Q1	15/3 Q5	+0.029 (+4.5)	32
Up	42d	Q5	5/5 Q1	-0.034 (-4.5)	38

- Fast ROC is close to independent of the level. Its quintile corners hold a balanced 27–38 names a day, so stacking the two adds information.
- Slow ROC (50/20) is the level again: same-direction corners hold 73–93 names, opposite corners 2–4. Its combination scores near the level alone and fades in 2022–26.
- Level + ROC 5/5: bigger spread than the level alone in 3 of 3 periods. Period t +2.4, +2.5, +1.5 vs +2.1, +2.7, +1.0. Corners hold 27–38 names a day vs 147–172 for a level quintile, so t gains less than the spread. The biggest gain is on up days.

5. Robustness — 20-session window, 2009–26, Q5–Q1

Variant — Down days	7d	14d	21d	42d
Baseline	-0.025 (-4.2)	-0.030 (-3.9)	-0.032 (-3.7)	-0.045 (-4.7)
Raw volume (no cleaning)	-0.025 (-4.2)	-0.030 (-3.9)	-0.032 (-3.7)	-0.045 (-4.7)
CV of volume levels	-0.023 (-3.5)	-0.030 (-3.6)	-0.033 (-3.3)	-0.053 (-4.8)
Rank lookback 126	-0.021 (-3.7)	-0.026 (-3.5)	-0.027 (-3.1)	-0.038 (-3.8)
Rank lookback 504	-0.022 (-3.7)	-0.029 (-3.7)	-0.030 (-3.4)	-0.042 (-4.3)
No history minimum	-0.023 (-3.9)	-0.027 (-3.6)	-0.029 (-3.4)	-0.039 (-4.2)
Ranked vs universe daily	-0.022 (-2.8)	-0.023 (-2.2)	-0.024 (-2.1)	-0.035 (-2.5)
Variant — Up days	7d	14d	21d	42d
Baseline	-0.011 (-1.8)	-0.019 (-2.4)	-0.021 (-2.4)	-0.037 (-3.9)
Raw volume (no cleaning)	-0.011 (-1.8)	-0.019 (-2.4)	-0.021 (-2.4)	-0.037 (-3.9)
CV of volume levels	-0.015 (-2.1)	-0.023 (-2.7)	-0.026 (-2.7)	-0.047 (-4.5)
Rank lookback 126	-0.008 (-1.4)	-0.018 (-2.4)	-0.020 (-2.4)	-0.034 (-3.7)
Rank lookback 504	-0.008 (-1.3)	-0.017 (-2.2)	-0.021 (-2.4)	-0.038 (-3.8)
No history minimum	-0.007 (-1.2)	-0.014 (-1.8)	-0.016 (-1.9)	-0.032 (-3.4)
Ranked vs universe daily	-0.011 (-1.4)	-0.020 (-2.0)	-0.022 (-1.8)	-0.035 (-2.5)

Other windows, full backtest (own-history rank)

Window — Down days	7d	14d	21d	42d
20 sessions	-0.025 (-4.2)	-0.030 (-3.9)	-0.032 (-3.7)	-0.045 (-4.7)
50 sessions	-0.016 (-3.3)	-0.023 (-3.6)	-0.025 (-3.3)	-0.028 (-3.3)
60 sessions	-0.010 (-2.2)	-0.017 (-3.1)	-0.017 (-2.7)	-0.020 (-2.4)
120 sessions	-0.008 (-1.9)	-0.012 (-2.3)	-0.013 (-2.1)	-0.016 (-1.9)
252 sessions	-0.002 (-0.4)	-0.005 (-0.9)	-0.004 (-0.7)	-0.010 (-1.3)
60 sessions, CV	-0.010 (-2.3)	-0.014 (-2.6)	-0.016 (-2.8)	-0.024 (-3.2)
Window — Up days	7d	14d	21d	42d
20 sessions	-0.011 (-1.8)	-0.019 (-2.4)	-0.021 (-2.4)	-0.037 (-3.9)
50 sessions	-0.006 (-1.3)	-0.014 (-2.2)	-0.018 (-2.4)	-0.022 (-2.6)
60 sessions	-0.009 (-2.1)	-0.014 (-2.4)	-0.015 (-2.2)	-0.017 (-1.9)
120 sessions	-0.006 (-1.5)	-0.011 (-2.0)	-0.013 (-2.1)	-0.017 (-1.9)
252 sessions	-0.003 (-0.9)	-0.005 (-0.9)	-0.009 (-1.4)	-0.015 (-1.8)
60 sessions, CV	-0.013 (-2.9)	-0.016 (-2.8)	-0.022 (-3.4)	-0.032 (-3.9)

Ticker by ticker (20-session window)

Per ticker	Dn 7d	Dn 14d	Dn 21d	Dn 42d	Up 7d	Up 14d	Up 21d	Up 42d
% tickers Q5 < Q1	55%	55%	54%	57%	50%	51%	52%	55%
Median IC	-0.006	-0.006	-0.007	-0.010	+0.001	-0.002	-0.002	-0.009

- Data cleaning moves no window's t by more than 0.08.
- Each robustness variant keeps the sign. Ranking each ticker against its own history beats ranking against the universe each day.
- Ticker by ticker the edge is thin: 2,155 tickers, median IC near zero. It shows up across the list as a group, not as a timing signal for one name.

6. What changed from earlier rounds

- Round 1 (2016–26, 20/10): volume volatility negative, ROC weak. Holds.
- Round 2 (2016–26 sweep to 60): recommended 40–60. Reversed. That result came from 2022–26. Over 2009–26, 20 is the strongest and long windows fail in 2009–16.
- Round 3 (2018–26, out-of-sample split): dropped ROC. Refined. Slow ROC with the level held fixed worked for 2009–2022 and has been flat since mid-2022. Fast ROC (5–15 / 3–5) is a separate, positive signal that holds in all three periods.
- Round 5 (2009–26 combinations): the 20-session level plus fast ROC widens the spread over the level alone in every period. Slow ROC adds little beyond the level.
- Fixed along the way: the risk-adjustment volatility was tied to the volume-volatility window (now fixed at 20), and bad volume prints are now cleaned. Neither changed the 20-session results.

Survivorship: the universe is today's list run backward, so names that delisted before 2026 are absent. All figures are relative to that universe.

Code and every CSV behind this report: [research/volume_volatility/](#) on branch [claude/volume-volatility-backtest-sjbtja](#).